



Time-varying persistence of house price growth: The role of expectations and credit supply[☆]

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ABSTRACT

House price growth in the U.S. has long been known to be highly persistent, but it is unclear whether this persistence itself varies over time and what drives such variation. We document pronounced time variation in the persistence of house price growth at both the national and metropolitan levels over the past four decades. Persistence in national house price growth rose sharply from the mid-1990s onward, and this increase was largely driven by historically less persistent regions in the South and Midwest. To study the drivers of these patterns, we estimate time-varying persistence measures and relate them to city-level proxies for credit supply and extrapolative expectations in panel regressions. We find that increases in house price persistence are systematically associated with local credit supply expansions, measured by per capita deposit growth, and with stronger extrapolative expectations, proxied by house price gaps relative to trend. These relationships remain robust when we instrument credit supply using a Bartik style shift-share measure based on national deposit growth.

Scope: Despite widespread agreement that U.S. house price growth is highly persistent, existing work has yet systematically examined how this persistence evolves over time, how it varies across cities and regions, or what economic forces drive these patterns. We fill this gap by documenting time-varying and cross-sectional patterns in house price persistence and by relating them to city-level measures of credit supply and extrapolative expectations.

Two-stage strategy: Our empirical strategy has two stages. In the first stage, we estimate time-varying persistence of house price growth for the national index and for 279 MSAs using a time-varying parameter autoregressive model with stochastic volatility. This delivers a panel

of persistence measures that captures both temporal and geographic variation in housing dynamics. In the second stage, we relate these persistence measures to city-level measures of credit supply, extrapolative expectations, and housing supply constraints in panel regressions, and assess robustness using Bartik-style instruments for exogenous shifts in local credit supply.

“Home prices have a lot of inertia and momentum, which makes housing markets more forecastable than stock markets.”

–Robert Shiller, at the AREUEA Policy Forum on Nov 16, 2022.

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1. Introduction

As highlighted in the opening quote, house price growth in the U.S. is known to be highly persistent.¹ If house prices rise rapidly in the current period, they tend to continue rising in the subsequent period, making current price growth a useful predictor of future house price movements. Since the seminal work of [Case and Shiller \(1989\)](#), a large literature has documented high persistence in U.S. house price growth, regardless of location, time period, or price index measurement (e.g., [Ghysels et al., 2013](#); [Guren, 2018](#); see further discussion in Section 2).² Yet most existing studies treat persistence as stable over time. Given the substantial structural changes in U.S. housing and mortgage markets over the past four decades, this assumption is restrictive and potentially misleading.

In this paper, we show that house price persistence is itself time-varying and that understanding this variation is crucial for interpreting housing cycles and its influence on key macroeconomic outcomes. [Fig. 1](#) displays the persistence of the FHFA's U.S. national house price growth since the early 1980s, measured by the sum of autoregressive coefficients (SARC) from autoregressive (AR) models estimated over 12-year rolling windows.³ The graph shows a nontrivial variation over time in the persistence of U.S. national house price growth. The SARC estimate began to rise in the mid-1990s, reached its peak in the early 2000s, and subsequently remained relatively high thereafter. Notably, the significant rise in persistence occurred before the onset of the 2000s housing cycle. The apparent time-varying behavior of persistence suggests that estimating it without allowing for parameter instability could be inaccurate, and the often cited high persistence of house price growth in the literature could be due to ignoring this temporal variation.⁴

Understanding time variation in house price persistence matters for several key reasons. First, house prices shape major macroeconomic outcomes, such as consumption and monetary policy transmission, so changes in persistence affect how housing markets influence the broader economy (e.g., [Choi and Jo, 2026](#)). Second, persistence determines how long housing shocks last. If persistence varies over time, the duration and strength of housing cycles also change, complicating monetary policy responses. Third, high and time-varying persistence can signal prolonged deviations from fundamentals, making housing bubbles and their timing harder to detect and predict (e.g., [Phillips et al., 2015](#)). Moreover, persistent house price growth at the regional level can amplify geographic disparities, widening cross-region price gaps and contributing to inequality through wealth and consumption channels.

¹ In the literature, 'persistence' is often used interchangeably with 'momentum' and 'serial (auto)correlation'. In this paper, we adopt the term *persistence* for consistency. See [Ghysels et al. \(2013\)](#) for a comprehensive discussion of persistence in house price growth.

² For example, [Ghysels et al. \(2013\)](#) document strong serial correlation in growth rates of widely used national house price indices, reporting values as high as 0.94 for the Case–Shiller index and 0.76 for the Federal Housing Finance Agency (FHFA) monthly house price index (HPI). U.S. urban house price growth similarly exhibits substantial persistence, with serial correlation around 0.7 ([Schindler, 2013](#)).

³ As discussed in [Appendix A](#), the serial autocorrelation coefficient (SARC) for national house price growth is estimated using [Hansen's \(1999\)](#) grid-bootstrap median-unbiased (MUB) estimator. The lag length is chosen according to the BIC criterion, allowing up to eight lags, and thus may vary across subsample periods. The horizontal axis in [Fig. 1](#) indicates the starting year of each 12-year rolling window. For example, 1995 corresponds to the subsample spanning 1995–2006. [Appendix A](#) provides additional details on the SARC estimation.

⁴ Structural shifts, or variations over time in mean or persistence, are known to induce an upward bias on persistence estimates obtained from standard AR models assuming a constant mean (e.g., [Ang and Timmermann, 2012](#); [Choi and Moh, 2007](#)).

Table 1

Diverse sources of persistence of house price growth in the literature.

Information costs	Frictions in demand	Supply costs and constraints
Asymmetric information	Business cycle	Housing market regulations
High transaction costs	Credit constraints	Long lag of construction
Non-standardized and heterogeneous properties	Extrapolative expectations	Construction costs
Illiquidity	Investors' sentiment	Geographic constraints
Search/gradual learning		

We make two contributions here. First, we document pronounced time variation in the persistence of U.S. house price growth at both national and metropolitan levels. While the visual evidence in [Fig. 1](#) is suggestive, it may not be sufficient on its own to establish time-varying persistence as an empirical regularity. We therefore provide formal tests of time variation in persistence and investigate its drivers through comprehensive regression analyses by examining movements in U.S. national and urban house prices over the past four decades.

Second, we examine the economic forces behind this time variation. House prices are in principle determined by the interplay of housing supply and demand, so the magnitude of house price growth persistence is likely affected by the persistence of demand shocks as well as the elasticity of supply response (see [Table 1](#) for various factors discussed in the literature). Guided by recent research highlighting the roles of beliefs and credit conditions in housing cycles (e.g., [Glaeser and Nathanson, 2017](#); [Favara and Imbs, 2015](#)), we focus on two demand-side factors that have increasingly been recognized as influential in the housing boom-bust cycle of the 2000s: extrapolative expectations and credit supply expansion (e.g., [Cox and Ludvigson, 2021](#); [Griffin et al., 2021](#)). This is because demand-side factors are widely recognized as the primary drivers of house price movements, given the slow adjustment of housing supply. Moreover, the literature widely documents that conventional demand fundamentals, such as income and population density, cannot fully explain persistent house price movements, particularly during the 2000s housing cycle (e.g., [Head et al., 2014](#)).

In our empirical analysis, we adopt a two-step approach, following [Gorodnichenko and Talavera \(2017\)](#), to study how demand-side factors relate to time-varying persistence in house price growth. In the first stage, we estimate house price persistence for 279 MSAs using a time-varying parameter autoregressive model with stochastic volatility (TVP-SV-AR model; e.g., [Primiceri, 2005](#)). We find that persistence rose markedly since the mid-1990s and that this increase was especially strong in cities in the South and Midwest that historically exhibited low persistence and slower house price growth. These findings indicate that persistence is not a fixed structural feature of housing markets; instead it evolves with market conditions and differs across locations.

In the second stage, we relate these persistence estimates to city-level demand characteristics in panel regressions, finding that our results are consistent with both the credit supply channel, measured by per capita deposit growth, and with the extrapolative expectation channel, proxied by house price gaps relative to trend. Instrumental-variable estimates based on Bartik-style credit supply shocks support a causal interpretation for the credit channel. Overall, the results indicate that changes in expectation formation and financial conditions help explain the change in persistence of house price dynamics over time.

The remainder of the paper is organized as follows. Section 2 reviews the recent literature on the determinants of house price persistence. Section 3 describes the data and key summary statistics. Section 4 introduces the TVP-SV-AR model and presents national-level persistence estimates. Section 5 extends the analysis to 279 MSAs,

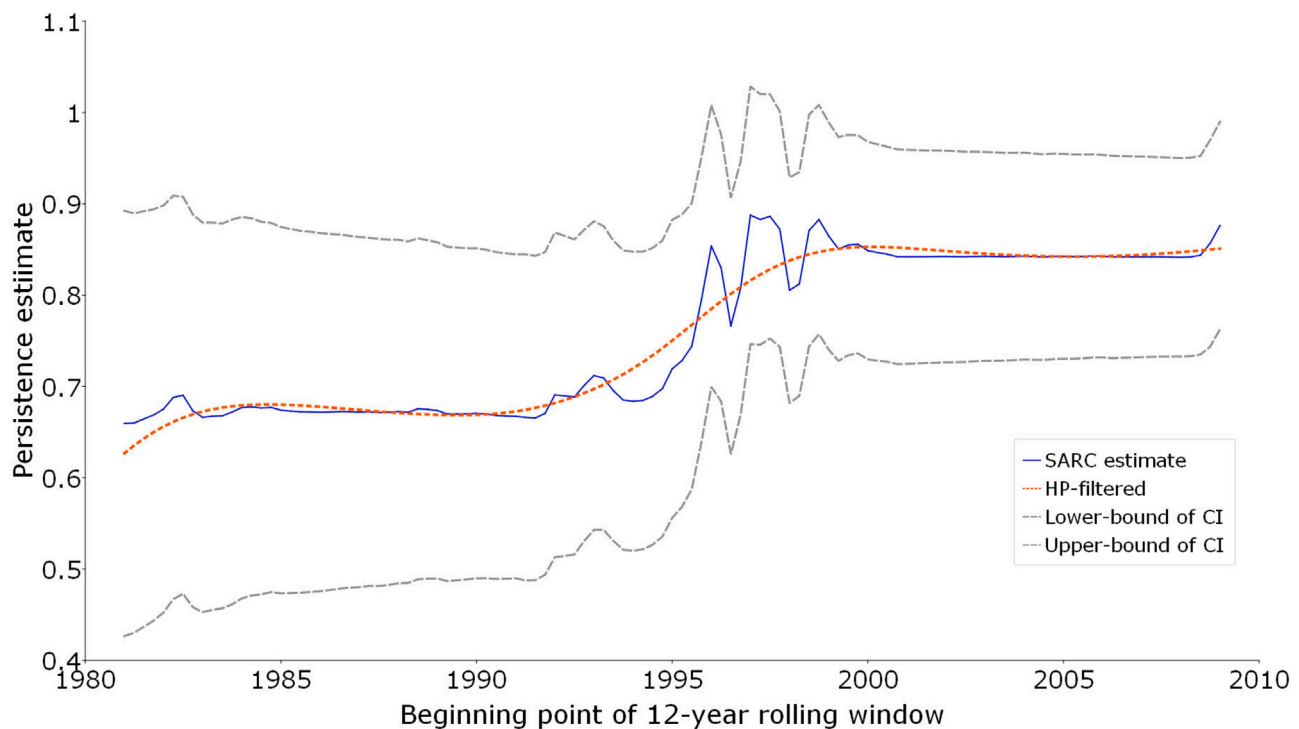


Fig. 1. Evolution of the SARC estimates with 95-percent confidence intervals for national house price growth using 12-year rolling windows.

Notes: This figure plots the SARC and its 95% confidence intervals for U.S. national house price growth, estimated from an AR model using 12-year rolling windows and Hansen's grid bootstrap median-unbiased (MUB) estimator. The horizontal axis indicates the starting year of each rolling window.

Table 2
Summary of selected previous studies on the persistence of urban HP growth.

Study	Data	Methodology	Driving force and notable findings
Beracha and Skiba (2011)	380 U.S. MSAs during 1983–2008	AR model & Dynamic panel GMM	Overconfidence of buyers. Stronger momentum in the West region and during 2004–2008.
Capozza et al. (2004)	62 U.S. MSAs during 1979–1995	Error correction model (ECM)	Information & supply costs. Higher serial correlation in areas with higher incomes, population growth and real construction costs
Griffin et al. (2021)	3725 U.S. zip code areas during 2003–2006	One-way fixed effect (FE) models	Housing market boom-bust in the 2000s is better explained by credit supply than extrapolative expectations
Gu (2002)	Quarterly CMHPI for U.S. states during 1975–1999	Moving average model and variance ratio test	Trading strategies of buyers & sellers. Marked difference in autocorrelation across locations and over time
Guren (2018)	MLS listing data during 2008–2013	IV analysis	Momentum is amplified by strategic complementarity of sellers
Gyourko and Voith (1992)	56 U.S. MSAs during 1971–1989	Two-way fixed effects model	No mention on driving force. Unequal persistence across MSAs
Lai and Order (2017)	45 U.S. MSAs during 1980–2012	Pooled Mean Group & Mean Group estimation	Stronger momentum in cities with lower housing supply elasticities
Oikarinen et al. (2018)	70 U.S. MSAs during 1980–2015	Two-stage regression with Pooled OLS & FMOLS regressions	Geographic differences in the momentum depends on housing supply elasticity
Plazzi et al. (2010)	53 U.S. MSAs during 1994–2003	GMM estimation	Commercial real estate returns are persistent and time-varying
Schindler (2013)	20 U.S. MSAs during 1987–2009	Autocorrelation & variance ratio tests	Transaction costs. Geographic heterogeneity in momentum
Sinai (2012)	249 U.S. MSAs during 1980–2010	Basic statistics	Non-fundamental factors. Heterogeneity in serial correlation
Titman et al. (2014)	97 U.S. MSAs during 1980–2011	Dynamic panel regression	Demand persistence & supply rigidities. Heterogeneity in the serial correlation

examining how credit supply and extrapolative expectations relate to persistence. This section also addresses endogeneity using city-level Bartik instruments and studies how housing supply constraints shape geographic heterogeneity in these effects. Section 6 concludes. The Appendix provides details on the persistence measure (SARC), the housing supply constraint measure, additional TVP-SV-AR results, and a discussion of popular panel regression approaches. The Online Appendix reports supplementary regressions using alternative measures of extrapolative expectations across different horizons.

2. Related literature

The persistence of house price growth has been extensively studied (see Table 2 for a partial list). However, existing evidence remains fragmented, largely because studies differ in their sample periods and empirical methodologies, leaving the underlying drivers of persistence an open question. Moreover, previous studies have neither systematically evaluated whether persistence is stable over time nor provided explanations for its time-varying behavior.

In theory, changes in house prices should reflect changes in both housing supply and demand. Therefore, the persistence of house price growth can be influenced by housing market characteristics linked to demand (e.g., business cycle, income, and population) and supply (e.g., regulation and geographic constraints), as well as basic market features (e.g., illiquidity, information and transaction costs, and the nonstandardized nature of property), as summarized in Table 1. Due to the slow adjustments of housing supply factors, however, demand factors are often viewed as the primary driver of house price movements (Head et al., 2014).⁵ While the literature suggests demand-side factors predominantly drive U.S. house price changes, the specific factors responsible for time variation in persistence remain unclear. Moreover, empirical evidence indicates that conventional demand fundamentals, such as income and population density, are insufficient alone to explain house price fluctuations, especially during the boom-bust cycle (e.g., Case and Shiller, 1989; Capozza et al., 2004).⁶

Recent research highlights two demand-side factors as key drivers of the 2000s housing boom-bust cycle: extrapolative expectations and credit supply expansion (e.g., Cox and Ludvigson, 2021; Kuchler et al., 2023).⁷ While debate continues over which of these factors played a more dominant role (e.g., Griffin et al., 2021; Howard and Liebersohn, 2023), examining their relevance for the house price persistence observed in our data is a worthwhile endeavor.⁸

⁵ Housing supply constraints – such as regulations, restrictive planning laws, and limited available land – affect price persistence by creating a time lag between demand shifts and supply adjustments. Even when demand factors are not persistent, they can still cause prolonged price changes if supply fails to respond in a timely manner.

⁶ The movements of fundamentals, such as income and population density, are neither large enough to account for the observed time variation in house price growth persistence nor sufficient to explain its geographic heterogeneity. For a contrasting perspective, see Chodorow-Reich et al. (2021), who attribute the 2000s boom-bust-rebound largely to fundamentals, though their focus is on price cycles rather than persistence. Additional results (not reported for brevity) show that time-varying persistence in income growth is at most weakly correlated with that of house price growth at both the national and MSA levels. These results are available upon request.

⁷ As summarized in Table 1, beyond the two factors we examine, the literature points to several other mechanisms behind the high persistence of house price growth, including search and information frictions (e.g., Head et al., 2014), heterogeneous beliefs and momentum trading (e.g., Piazzesi and Schneider, 2009), and strategic complementarities (e.g., Guren, 2018). While potentially important, we do not incorporate these channels here, both because they are less directly relevant for aggregate (city-level) price dynamics and because suitable geographic-level measures are not readily available.

⁸ See Howard and Liebersohn (2023) for an overview of the ongoing debate over the roles of expectations versus credit in the boom-bust cycle.

Expectations are a likely driver of house price dynamics, as individuals often base their expectations for future prices on recent price trends. Extrapolative, backward-looking expectations are frequently cited as a potential cause of price persistence (e.g., Case and Shiller, 1987; Glaeser and Nathanson, 2017; Granziera and Kozicki, 2015; Kuchler et al., 2023). In fact, numerous studies, including those based on surveys and lab experiments, support the role of extrapolative expectations (e.g., Case et al., 2012; Piazzesi and Schneider, 2009). Some recent studies also underscore the importance of expectations in explaining house price movements during the 2000s housing boom-bust cycle (e.g., Duca et al., 2021; Glaeser and Nathanson, 2017; Kuchler et al., 2023, among others). However, Granziera and Kozicki (2015) show that in an asset-pricing model with rational expectations, extrapolative expectations fail to reproduce the observed trajectory of U.S. house prices.

While the assumption of extrapolative expectations is widely accepted, some skepticism remains regarding its relevance to house price persistence for several reasons. First, establishing a causal link between expectations and house price growth persistence is challenging, as extrapolative expectations could simply be a result of persistent price movements rather than a cause. Second, expectations may play a smaller role in price growth persistence when income or credit constraints limit buyers' ability to act on their expectations of future price movements. Additionally, the availability of reliable measures of expectations poses another challenge. Surveys are often used to gauge expectations (e.g., Kuchler et al., 2023), but they are limited to specific geographic areas and time periods. As an alternative, past house price growth is frequently used in empirical research as a non-survey proxy for expectations, reflecting a backward-looking perspective. For example, Duca et al. (2021) recommend using annualized house price appreciation over the previous four years as a proxy for such expectations.⁹ In this study, we proxy extrapolative expectations using the house price gap, defined as the deviation of observed house prices from their long-run trend, where the trend is estimated using the regression-based filter proposed by Hamilton (2018). The underlying intuition is that expectations about future house price growth are shaped by departures from long-run norms rather than by short-run fluctuations alone. Large positive deviations from trend characterize environments in which recent price growth is unusually strong and salient, making past appreciation more likely to be extrapolated into the future. For this reason, the house price gap provides a more informative proxy for extrapolative expectations than simple measures of lagged house price growth.¹⁰

Changes in credit conditions are also frequently considered a key factor in the housing boom of the 2000s. The idea is that expanded access to mortgage credit boosted housing demand, leading to rising prices and greater market persistence (Favara and Imbs, 2015; Favilukis et al., 2017; Mian and Sufi, 2009). In comparing the influence of credit supply with that of extrapolative expectations, Griffin et al. (2021)

Although these factors often interact and are difficult to disentangle, the geographic analysis in this study may yield testable insights into the competing explanations.

⁹ Recent literature introduces additional non-survey measures of housing market expectations, such as media-based sentiment (Soo, 2018). Unfortunately, these measures are not available for the extensive geographic regions studied here. For a detailed review of housing market expectations, see Kuchler et al. (2023).

¹⁰ We construct house price gaps using the Hamilton (2018) filter over a two-year horizon. As documented in Appendix B, however, the regression results based on Hamilton-filtered house price gaps are sensitive to the choice of horizon. In panel regression analyses, the two-year house price gap yields coefficients with the anticipated positive sign, consistent with the extrapolative expectations channel, whereas the four-year and eight-year gaps produce counterintuitive negative estimates. In light of this sensitivity, we focus on the two-year house price gap throughout the paper.

contend that the surge in housing demand during the 2000s boom-bust cycle was largely driven by credit supply, rather than expectations.

Measuring exogenous shifts in credit supply, however, poses some challenges for empirical analysis, as credit supply is often intertwined with other factors like housing speculation. Moreover, there is no consensus on the sources behind credit supply expansion. Some studies attribute the expansion in credit supply and the subsequent housing boom to the widespread use of mortgage securitization (e.g., [Loutskina and Strahan, 2009](#); [Mian and Sufi, 2009, 2022](#)), while others point to the role of banking deregulation during the 2000s housing boom (e.g., [Favara and Imbs, 2015](#)). Since borrowing capacity depends on credit availability, banking integration through deregulation likely played a key role in expanding credit supply and shaping house price dynamics. [Favara and Imbs \(2015\)](#) demonstrate that banks' ability to open branches across state lines had a significant impact on the 2000s housing boom, primarily by expanding credit availability.¹¹ In this study, we use proxy measures of credit supply related to banking integration, drawing on data from bank deposits.

3. The data

The data for this study were collected from a variety of publicly accessible sources. Specifically, we retrieved *quarterly* Housing Price Indices (All-Transactions Indexes) for the US at the national level as well as for numerous Metropolitan Statistical Areas (MSAs) from the Federal Housing Finance Agency (FHFA). The FHFA HPI reflects a weighted average of price changes in single-family homes financed with conforming loans, which have been sold or refinanced multiple times. A key advantage of the FHFA HPI dataset is its broader geographical coverage compared to other popularly used datasets in the literature, such as the S&P CoreLogic Case-Shiller HPI, which only covers 20 metropolitan areas. Another significant strength of the FHFA dataset is its long data span, which stands out particularly when compared to datasets like Zillow and CoreLogic. The relatively long data span is crucial for accurately estimating persistence. It enables the construction of a large and comprehensive panel, which is valuable for examining the time-varying feature of persistence.¹² The original HPI data were obtained from the FHFA website (<https://www.fhfa.gov>) and seasonally adjusted using the U.S. Census Bureau's X-13 ARIMA methodology. This adjustment allows us to compute seasonally adjusted quarter-over-quarter house price growth rates.

From a pool of more than 300 MSAs available throughout the sample period, we selected 279 metropolitan areas for which all key variables required for the empirical analysis are available. The resulting city-level dataset consists of quarterly observations for 279 MSAs spanning 1980:Q1 to 2021:Q3 ($T = 167$, $N = 279$). This long panel encompasses multiple boom-and-bust cycles in the U.S. housing market. Moreover, the sample provides extensive geographic coverage, representing over 80% of the U.S. population throughout the study period. [Fig. 2](#) displays a map of the 279 MSAs included in our sample, where larger circles indicate greater increases in house price growth persistence after 1997, as measured by the SARC estimates. The selection of 1997 as the breakpoint is motivated by the visual patterns documented in [Fig. 1](#). It is notable that many MSAs with signifi-

cant increases in persistence are located in the Midwest and Southern states.

To gain further insights into the changes in persistence over time and across cities, [Fig. 3](#) illustrates the relationship between SARC estimates for the pre-1997 period (1980–1997, horizontal axis) and the post-1997 period (1998–2020, vertical axis) across the 279 MSAs in our sample. As shown in [Fig. 3](#), persistence appears to have increased in many MSAs after 1997. Again, a substantial increase in persistence is evident in MSAs that formerly exhibited relatively low persistence, especially those in the Southern states, consistent with the geographic patterns documented in [Fig. 2](#).

Other MSA-level variables employed in this study are sourced from various publicly-available datasets. Per capita bank deposits are obtained from the Summary of Deposit (SOD) data, available at the FDIC website (<http://www7.fdic.gov/sod/>). The SOD dataset provides annual observations (as of June 30 of each year) on all deposit-insured commercial banks and savings associations, regarding the ownership, location, and deposits of each U.S. bank branch from 1994 onwards. This restricts the MSA-level regression analysis to the post-1994 period. Data on per capita income and population density are obtained from the Bureau of Economic Analysis (BEA) Regional Economic Accounts and are available annually for each MSA in the sample. Unemployment rate data are collected from the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) program.

For the measure of housing supply constraints, we employ the MSA-level Land Unavailability (LU) index developed by [Lutz and Sand \(2023\)](#), obtained from the authors' webpage.¹³ As described in [Appendix C](#), the LU index captures the time-varying proportion of undevelopable land at the MSA level. As an alternative proxy for housing supply constraints, we also utilize the Wharton Residential Land Use Regulation Index (WRLURI), updated by [Gyourko et al. \(2021\)](#). This time-invariant, cross-sectional index provides a comprehensive measure of local land-use regulatory stringency at the MSA level, with higher values indicating more restrictive regulatory environments.

4. Time-varying parameter autoregressive (TVP-SV-AR) model

While [Figs. 1](#) through [3](#) provide compelling evidence of time variation in the persistence of both national and urban house price growth, a more formal analysis is needed to substantiate this observation. Research in macroeconomics has shown that models with time-varying parameters (TVP) are particularly effective in capturing the dynamic behavior of key variables such as inflation (e.g., [Aastveit et al., 2023b](#); [Nakajima, 2011](#); [Primiceri, 2005](#)). Notably, the time-varying parameter vector autoregressive (TVP-VAR) model, originally proposed by [Primiceri \(2005\)](#), has been widely adopted for its flexibility and effectiveness in capturing the evolving dynamics of economic data.

Building on this approach, our study employs a time-varying parameter autoregressive model with stochastic volatility (hereafter, TVP-SV-AR model) to estimate the time variation in house price growth persistence. We focus on the TVP-SV-AR model instead of the TVP-SV-VAR model because our primary goal is to measure persistence within a univariate autoregressive framework. This model is well-suited for our analysis for two main reasons. First, as highlighted by [Leamer \(2007\)](#), the close relationship between the business and housing cycles makes the TVP model a reasonable choice for capturing potential nonlinearities in house price growth. Second, since house price is closely tied to the user cost of owner-occupied shelter, a major component of the U.S. Consumer Price Index (CPI), its growth shares conceptual similarities with inflation, whose dynamic behavior is well-modeled using TVP approaches.

¹¹ Banking deregulation expanded mortgage credit supply in two key ways: first, by allowing banks to collect more deposits and extend more loans by opening branches across state borders, and second, by loosening lending standards due to increased competition in local lending markets (e.g., [Choi and Hansz, 2021](#)).

¹² In the earlier part of our sample, smaller MSAs may have relatively few securitized transactions, which could affect the precision of the FHFA house price indices. As documented in the Online Appendix, our results are robust to this concern. We are grateful to an anonymous reviewer for drawing our attention to this issue.

¹³ <https://github.com/ChandlerLutz/LandUnavailabilityData>.

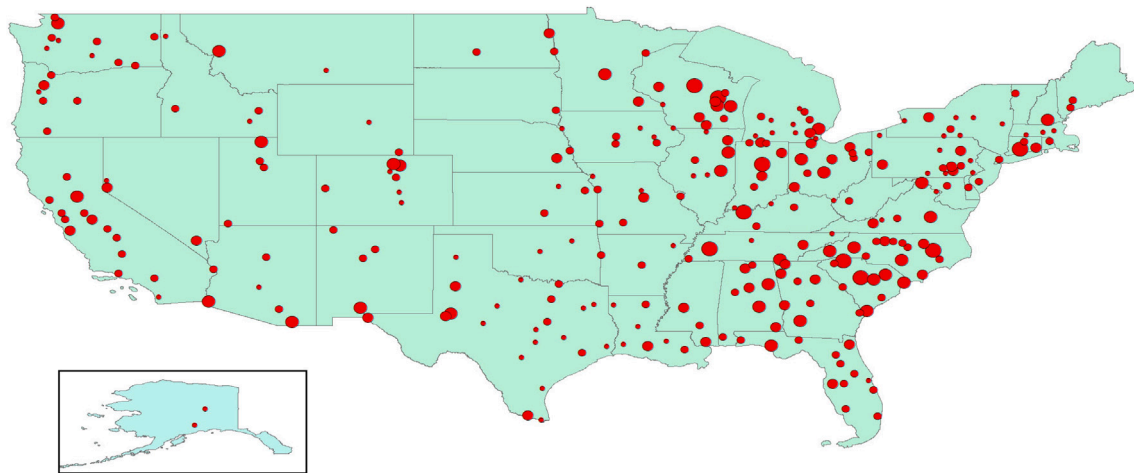


Fig. 2. Map of 279 MSAs based on the persistence change after 1997.

Notes: In this figure, larger circles represent greater increases in house price growth persistence after 1997, as measured by the SARC estimates.

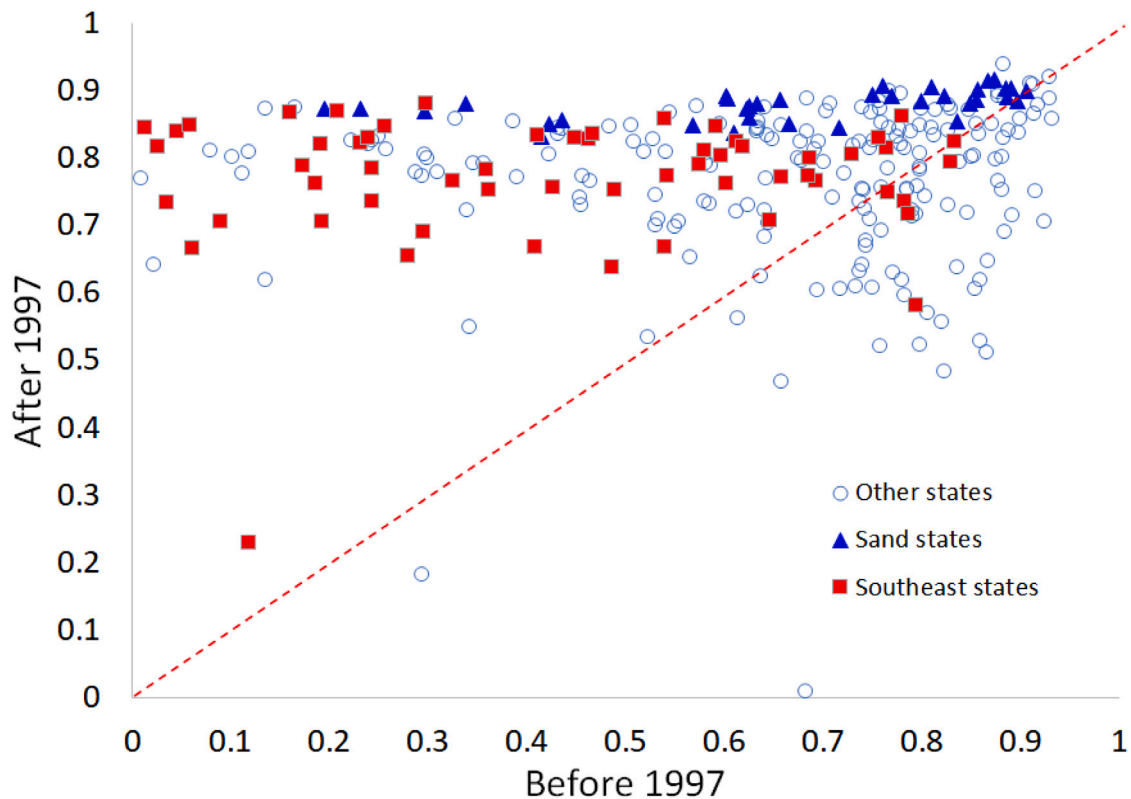


Fig. 3. Change in persistence of MSA house price growth before and after 1997.

Notes: This figure illustrates the association of SARC estimates across 279 MSAs, comparing the sample periods prior to 1997 (on the horizontal axis) and those following 1997 (on the vertical axis).

We incorporate stochastic volatility (SV) into the TVP framework to enhance estimation accuracy and reduce the risk of model misspecification (e.g., Nakajima, 2011). Integrating stochastic volatility with the TVP-AR model enables us to flexibly and jointly capture time variation in both persistence and volatility. In the absence of time-varying volatility, TVP models may yield biased parameter estimates (Primiceri, 2005). Accordingly, allowing for simultaneous time variation in both parameters and volatility is essential to avoid misleading inference (Chan, 2018; Lubik and Matthes, 2015). Additionally, the TVP-SV-AR model is capable of capturing nonlinearities arising from structural

economic shifts and heteroskedastic shocks, which are often observed in housing markets.¹⁴

To our knowledge, no previous research has employed the TVP-SV-AR model to estimate house price persistence, especially at the

¹⁴ Previous studies have shown that nonlinear models, such as Markov switching models, effectively capture key aspects of house price dynamics (e.g., Ang and Timmermann, 2012; Guirguis et al., 2005). However, Primiceri (2005) suggests that TVP models may better reflect shifts in aggregate private sector behavior compared to Markov switching models.

subnational level. Guirguis et al. (2005) applied a TVP-AR model to the U.S. national house price and found considerable parameter instability. Christou et al. (2019) also used a TVP-VAR model to study the impact of uncertainty shocks on the U.S. housing market. Relatedly, Aastveit et al. (2023b) utilized a structural VAR model with TVP and SV (TVP-SV-VAR) and found some evidence of time variation in the Fed's response to house prices. None of these studies, however, focus on the persistence of house price growth.

We consider the following TVP-SV-AR model,

$$y_t = \alpha_t + \rho_t y_{t-1} + \sum_{k=1}^{p-1} \zeta_{k,t} \Delta y_{t-k} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_t^2), \quad (1)$$

where y_t represents the house price growth in time t and σ_t^2 denotes the time-varying variance of housing price disturbances. All parameters in Eq. (1) are time varying such that $\Lambda_t = \{\alpha_t, \rho_t, \zeta_{1,t}, \dots, \zeta_{p-1,t}\}'$. Among them, the key parameter of interest, ρ_t , represents the SARC at time t and captures the time-varying persistence of house price growth. The optimal lag length ($p - 1$) is selected by the BIC rule with the maximum lag length set as eight. Following Primiceri (2005), the vector of coefficients (Λ_t) is assumed to evolve as random-walks, with associated disturbances that are jointly normally distributed and mutually and serially uncorrelated. Specifically,

$$\Lambda_{t+1} = \Lambda_t + u_{t+1}, \quad u_{t+1} \sim N(0, \Sigma). \quad (2)$$

This parsimonious specification allows persistence to have both permanent and purely transitory components (see Primiceri, 2005). The model can also capture numerous patterns without introducing additional parameters that need to be estimated (Lubik and Matthes, 2015).

Following Nakajima (2011), it is assumed that the variance of ε_t evolves according to $\sigma_t^2 = \gamma \exp(h_t)$, where the law of motion for h_t is given by,

$$h_{t+1} = \phi h_t + \eta_{t+1}, \quad \eta_{t+1} \sim N(0, \sigma_\eta^2). \quad (3)$$

In addition to the independent noise component, η_t , the SV model can capture the time variation of the variance of disturbances, where past volatility of house price growth can affect the current volatility of house price growth.¹⁵ As a result, the volatility of house price growth in the TVP-SV-AR model can stem either from changes in the transmission mechanism, reflected in the time-varying parameters of the mean equation, or from the impact of the exogenous shock η_t in Eq. (3) (Primiceri, 2005).¹⁶

While theoretically it might be possible to estimate the TVP-SV model with conventional methods based on a state-space representation, the computational burden for evaluating the likelihood function poses a significant challenge, particularly given the large number of MSAs we analyze. To address this, we use Bayesian inference, employing the Markov Chain Monte Carlo (MCMC) algorithm with a Gibbs sampler.¹⁷

¹⁵ In the literature, volatility of house price growth, which is unobservable by nature, is often modeled using conditional variance approaches like GARCH models (see Ghysels et al., 2013). While GARCH models explicitly constrain volatility to only be a function of past observations of the residuals of house price growth, the SV approach links variance to an independent stochastic component, allowing volatility to be influenced by new information. For this reason, SV is preferred (see Chan and Grant, 2016). Higgins and Sapci (2024) also show that the substantial time-varying volatility observed in U.S. house prices can be effectively captured using a stochastic volatility model.

¹⁶ A thorny issue with the TVP-SV-AR model is its tendency to attribute time variation in the data to the SV term rather than the lag coefficients. However, this is less problematic in our case, as our findings indicate that the fluctuations in house price growth are primarily driven by changes in the lag coefficients rather than the SV component.

Against this backdrop, we estimate the *univariate* TVP-SV-AR model using the national house price growth in Eq. (1). The results are displayed in Figs. 4 and 5 and highlight several key observations. First, all estimated coefficients show significant variation over time, especially the coefficient of interest, ρ_t . As shown in Fig. 4, the SARC estimate ($\hat{\rho}_t$) fluctuated around the value of 0.3 before the mid-1990s when it surged to around 0.8 in the early 2000s, similar to the visual evidence noted in Fig. 1. As illustrated in the figure, the dotted one-standard-deviation bands around the SARC estimates also rise sharply around the mid-1990s, closely tracking the dynamics of the point estimates. This suggests substantial time variation in the persistence of national house price growth. Moreover, the notable increase in persistence predates the housing market boom of the 2000s, suggesting that the rise in persistence was likely driven by other factors than the housing market boom itself.

Similar time-varying patterns are witnessed in other coefficient estimates. As illustrated in Fig. 5(a), the volatility of national house price growth remained relatively stable until the mid-1990s, after which it increased, peaking in the early 2000s before returning to pre-mid-1990s levels. This indicates that SV of national house price growth exhibits a hump-shaped profile around the early 2000s, possibly reflecting structural changes in the U.S. housing market triggered by shocks during that period. Interestingly, the time-varying volatility captured by the SV term, as depicted in Fig. 5(a), is less pronounced than the variation in persistence shown in Fig. 4. This suggests that the dynamics of national house price growth may have been driven more by time-varying persistence – potentially due to systematic policy changes or shifts in private sector behavior – rather than by the volatility of non-policy-related shocks. Fig. 5(b) plots the evolution of the intercept term ($\hat{\alpha}_t$) along with the corresponding one-standard deviation intervals. The estimated intercept ($\hat{\alpha}_t$) exhibits a countercyclical pattern relative to house price growth, declining during the housing boom and rising during the subsequent housing market downturn.

5. Factors behind the house price persistence

The results of our TVP-SV-AR model analysis in the preceding section confirm the substantial variation in the persistence of U.S. house price growth across different periods. It is equally important to comprehend the factors behind the temporal variation of the persistence. Since national house prices are an aggregate of numerous sub-national prices, and large persistence in overall house price growth can stem from aggregating a variety of local house prices (e.g., Choi et al., 2006; Choi and Choi, 2016), geographical considerations may provide insights into the causes of this house price persistence.

In this section, we explore the factors contributing to the temporal variation in house price growth persistence by analyzing geographically disaggregated data. According to the spatial equilibrium model (e.g., the Rosen-Roback model), spatial heterogeneity is inherent in house prices, as they are influenced by local factors such as wages, amenities, and area-specific housing supply conditions. As a result, the housing literature consistently highlights the significant variability in house price dynamics across different regions (Case and Shiller, 1989; Gyourko and Voith, 1992). Recent research, some of which is summarized in Table 2, has delved into this issue at a disaggregated level, uncovering significant geographic heterogeneity in the persistence of

¹⁷ The basic idea underlying the MCMC approach is that computational methods can be used to obtain moments from the desired posterior distribution by repeatedly sampling a Markov chain. Our Gibbs sampling algorithm incorporates the correction to the ordering of steps suggested by Del Negro and Primiceri (2015). To compute the posterior estimates, we use an initial burn-in of 100, followed by an additional 1000 draws. For further details on the implementation of the TVP-SV-AR model, see Nakajima (2011) and Lubik and Matthes (2015).

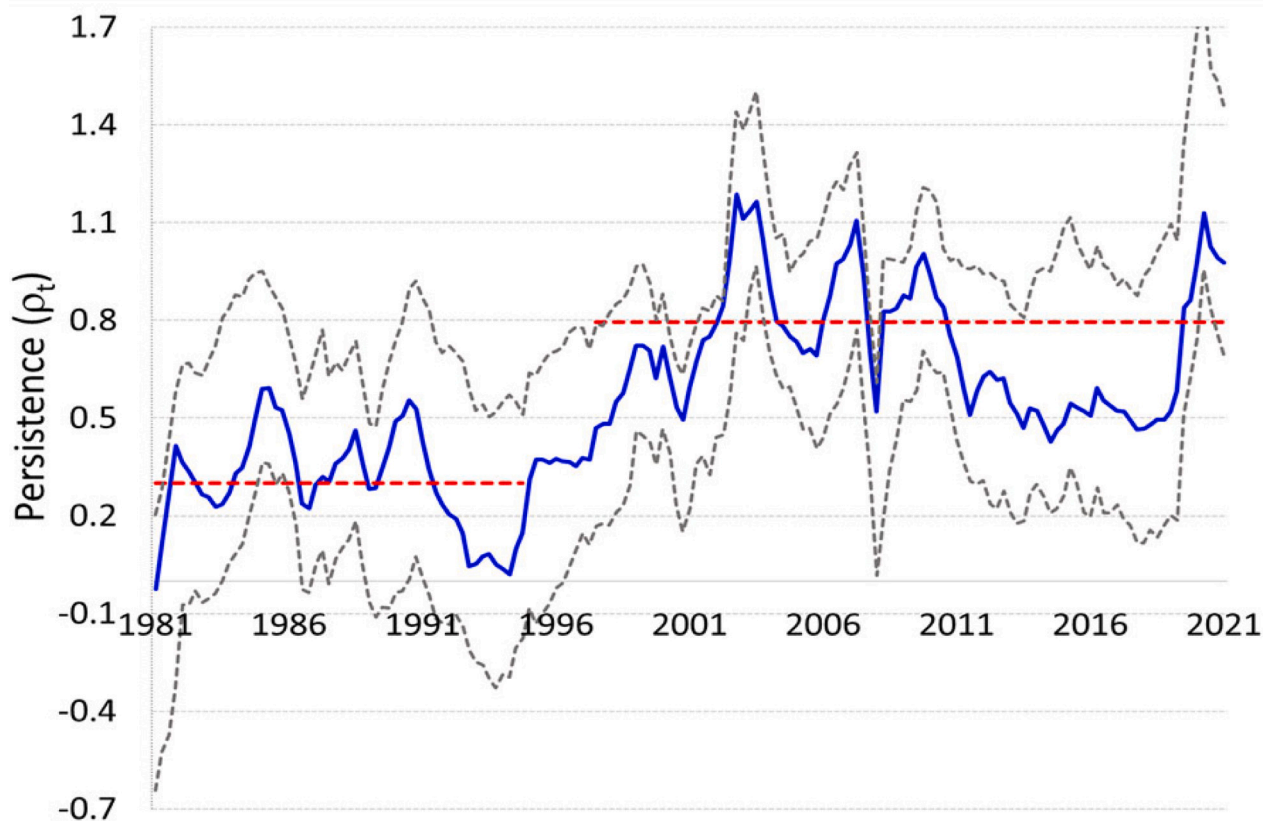


Fig. 4. Time-varying SAR coefficient (solid line) and one S.D. bands (dotted line) estimated in TVP-SV-AR model for the U.S. house price growth.

Notes: This figure presents the estimated time-varying persistence (ρ_t) of national house price growth based on the univariate TVP-SV-AR model specified in Eq. (1).

house price growth. This heterogeneity may arise from various factors, including demand-side influences and housing supply conditions (Capozza et al., 2004; Duca et al., 2021; Glaeser et al., 2008; Gyourko et al., 2008; Oikarinen et al., 2018; Saiz, 2010). Therefore, analyzing disaggregated data can help clarify the underlying drivers by linking geographically distinct patterns in house price growth persistence to a range of potential explanatory factors under scrutiny.

To this end, we focus on the wider variation in house price growth persistence observed in urban areas (e.g., Oikarinen et al., 2018). To be specific, we estimate the urban house price persistence using the univariate TVP-SV-AR model specified in Eq. (1), and then relate these persistence estimates to measures of credit supply, extrapolative expectations, and other control variables within a panel data regression framework.

Before proceeding, Fig. 6 plots the estimated values of ρ_{it} from the TVP-SV-AR model in Eq. (1) for 279 MSAs over the period from 1980 to 2020. The left axis represents the time dimension, while the right axis corresponds to the MSAs. The figure highlights substantial variation in the persistence of quarterly urban house price growth, not only over time but also across MSAs. To illustrate these dynamic patterns across regions, we selected four major MSAs, each representing one of the four U.S. Census regions: Atlanta (South), Chicago (Midwest), Los Angeles (West), and New York City (Northeast). Fig. 7 presents the persistence estimates ($\hat{\rho}_{it}$) for these four selected cities, which reveal several key insights. First, while house price growth persistence fluctuates over time in each city, the patterns of variation differ markedly across locations. The top two panels of Fig. 7, which display estimates for Atlanta and Chicago, show a sharp rise in persistence during the mid-1990s — from near zero (indicating no persistence) to approximately 0.8 (indicating high persistence). Notably, this trend closely resembles the pattern observed in Fig. 1 for U.S. national house price growth. In

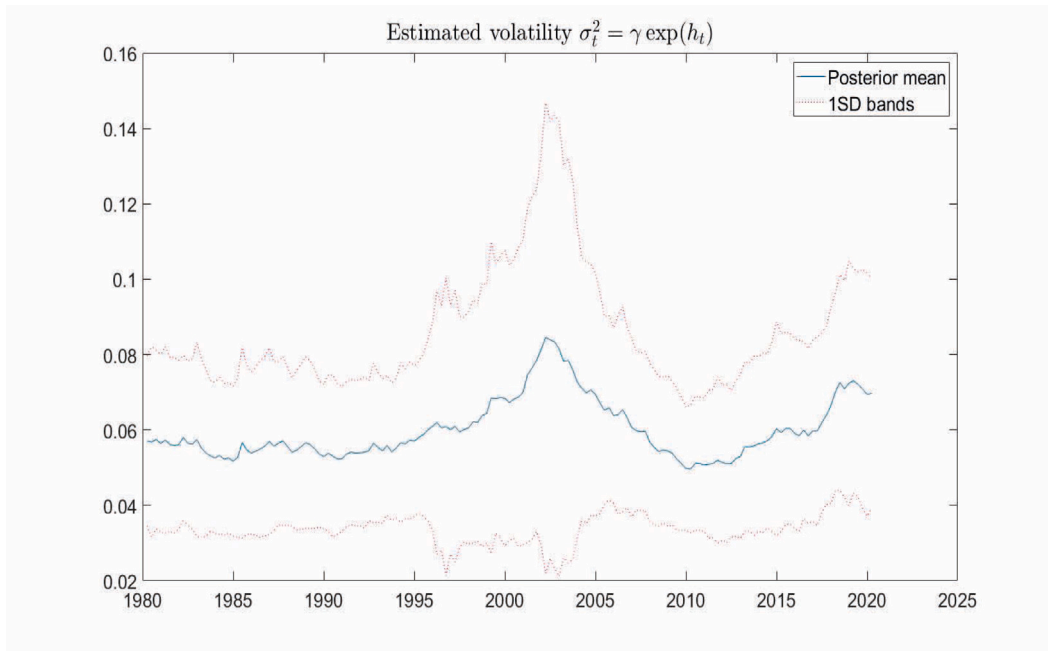
contrast, cities in the West and Northeast exhibit less dramatic movements. In Los Angeles, persistence remained consistently high, around 0.8, with only moderate fluctuations throughout the sample period. A similar pattern emerges in New York City, though with somewhat more pronounced variability compared to Los Angeles. Overall, these findings suggest that the house price persistence observed in national U.S. data may have been largely driven by cities in the South and Midwest rather than those in the Northeast and West — a conclusion further supported by the visual evidence in Figs. 2 and 3.

5.1. Panel data regression analyses with city-level data

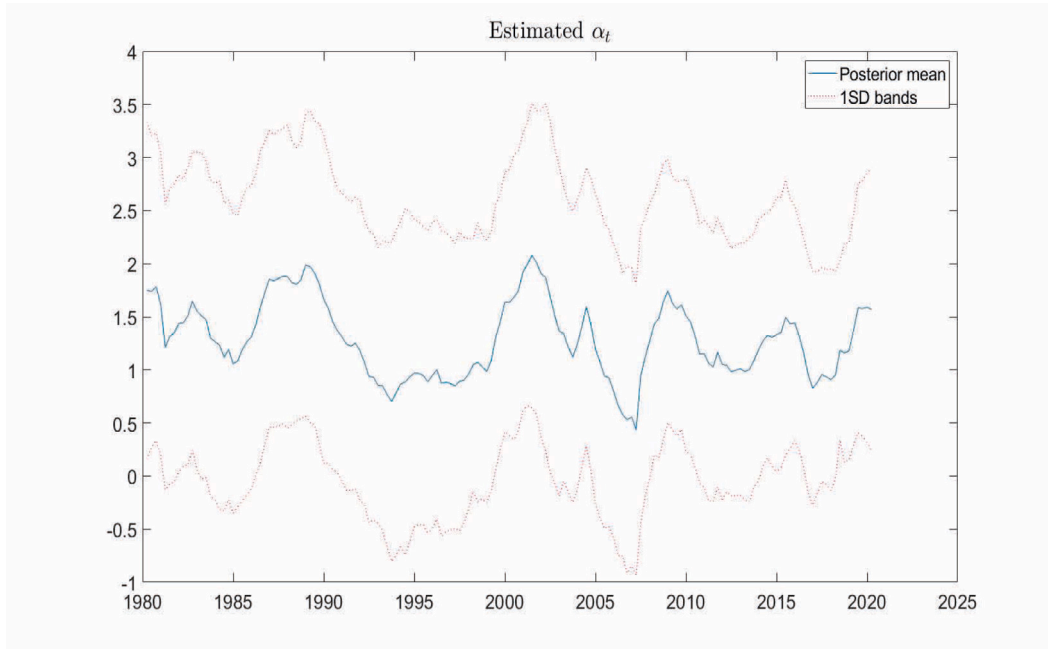
We next conduct panel regression analyses using city-level house price data for 279 U.S. MSAs ($N = 279$) over the period 1994–2020 ($T = 27$). Due to data availability constraints — most notably that key explanatory variables such as bank deposits are reported only annually beginning in 1994 — we conduct the panel data analysis using annual data. Accordingly, the city-level persistence estimates ($\hat{\rho}_{it}$) are annualized by averaging across the four quarterly observations.

We proceed in two sequential stages. First, we estimate the persistence of house price growth for each of the 279 MSAs individually using the univariate TVP-SV-AR model outlined in Eq. (1). In the second stage, we regress the estimated city-level persistence ($\hat{\rho}_{it}$) on city-level characteristics related to credit supply, expectations channels, and additional control variables within a panel data regression framework.¹⁸

¹⁸ This two-stage approach is similarly employed in influential studies like Gorodnichenko and Talavera (2017), Section 3), where the authors first estimate the speed of online price adjustment in the U.S. and Canada, and then use these estimates in subsequent regression analysis. A comparable approach is also followed by Chudik et al. (2021).



(a) Posterior estimates for stochastic volatility of national house price growth

(b) Posterior estimates for intercept term (α_t) for national house price growth**Fig. 5.** Posterior estimates of stochastic volatility (top) and intercept term (bottom) estimated in TVP-SV-AR model for national house price growth.

Notes: The top panel plots the volatility (σ_t^2) of national house price growth with its one-standard-deviation bands, estimated from the univariate TVP-SV-AR model in equation (1). The bottom panel shows the evolution of the intercept term ($\hat{\alpha}_t$) with corresponding one-standard-deviation bands from the same model.

The panel structure of the persistence estimates ($\hat{\rho}_{it}$) enables us to assess the impact of the two factors under comparison on changes in the persistence of urban house price growth, while controlling for unobserved MSA-specific characteristics, common macroeconomic shocks, and time-varying housing market conditions at the MSA level. Specifically, we apply two widely-used panel data techniques suited to our balanced panel dataset: the two-way fixed effects (TWFE) model and the Common Correlated Effects (CCE) estimation method developed by Pesaran (2006).

Our analysis is based on the following standard panel data model:

$$\hat{\rho}_{it} = \mu_i + \beta_1 CS_{it} + \beta_2 EXP_{it} + W'_{it}\varphi + v_{it}, \quad (4)$$

where $v_{it} = \delta_t + \epsilon_{it}$ (TWFE) or $v_{it} = \gamma'_i \mathbf{f}_t + \epsilon_{it}$ (CCE), with δ_t and $\mathbf{f}_t$ capturing aggregate macroeconomic shocks common to all localities. The dependent variable ($\hat{\rho}_{it}$) is the SARC estimate of house price growth in city i for year t , obtained from the TVP-SV-AR model in Eq. (1).

It is worth mentioning that the dependent variable ($\hat{\rho}_{it}$) is an ‘estimated’ variable, and therefore it is subject to sampling uncertainty. Let $\hat{\rho}_{it} = \rho_{it} + \omega_{it}$, where ω_{it} is the estimation error. Substituting this expression in Eq. (4), we obtain

$$\rho_{it} = \mu_i + \beta_1 CS_{it} + \beta_2 EXP_{it} + W'_{it}\varphi + v_{it}^*,$$

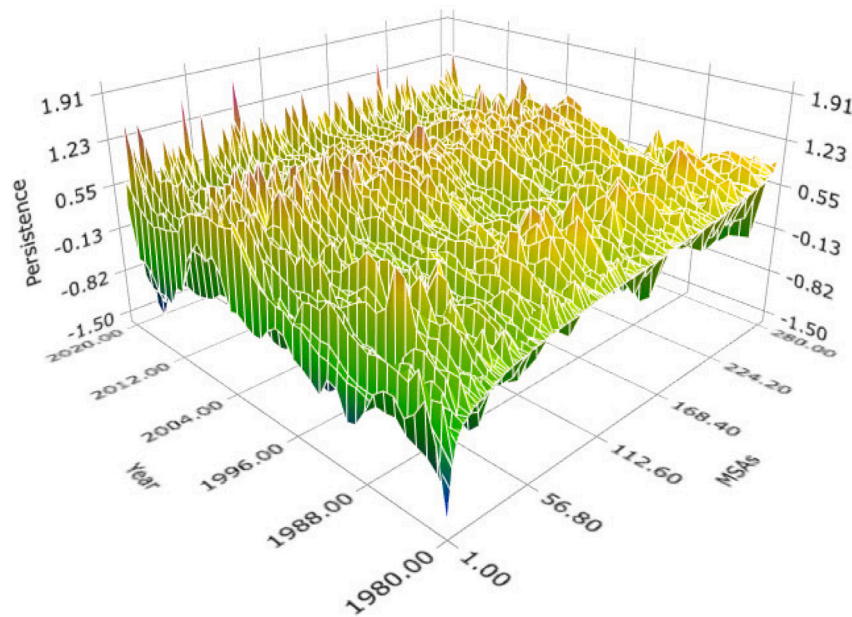


Fig. 6. Time-varying persistence estimated from TVP-SV-AR model across 279 MSAs over 1980–2020.

Notes: This figure plots the estimated $\hat{\rho}_{it}$ values from the TVP-SV-AR model in equation (1) for 279 MSAs over 1980–2020. Time is shown on the left axis, and MSAs are shown on the right axis.

where $v_{it}^* = v_{it} - \omega_{it}$ is composed of the original error (v_{it}) and the estimation error (ω_{it}). Thus, the correlation between the estimation error (ω_{it}) and our regression variables is crucial for the panel estimation of β_1 and β_2 . We assume $E(\omega_{it} | EXP, CS, W) = 0$, which ensures that the consistency and asymptotic normality of the TWFE and CCE estimators remain valid. In this case, ω_{it} only influences the panel estimates by increasing the standard errors (i.e., greater sampling uncertainty in β_1 and β_2), which can still be consistently estimated.

In Eq. (4), the key explanatory variables of interest are CS_{it} and EXP_{it} , which respectively represent credit supply and extrapolative expectations in city i for year t . City-level credit supply (CS_{it}) is proxied by *per capita deposit growth* in city i for year t , as deposit availability is a primary funding source for banks and thus a reliable indicator of local credit supply conditions (e.g., Aguirregabiria et al., 2020; Becker, 2007).¹⁹ Given the role of banking integration in expanding credit supply (e.g., Taşkın and Yaman, 2023), cities with stronger connections to the national banking system are likely to experience higher levels of credit supply.²⁰ As a proxy for city-level extrapolative

expectations (EXP_{it}), we use the *previous two-year house price gap* based on the Hamilton (2018) filter as detailed in Appendix B.²¹ Intuitively, MSAs with faster growth in either credit provision or extrapolative expectations are likely to see a larger increase in persistence. Therefore, we expect positive coefficients for both variables, i.e., $\beta_1 > 0$ and $\beta_2 > 0$.

W_{it} represents a set of time-varying control variables at the MSA level, including per capita income growth, population density growth, and the unemployment rate in city i for year t . By incorporating these time-varying city characteristics (W_{it}) as controls, we aim to reduce potential biases from omitted variables. Population density is linked to housing market transaction volumes, which affect information costs (e.g., Capozza et al., 2004). Unemployment rate is included to account for labor market frictions, which can influence housing demand – higher unemployment may reduce house prices and transaction volume by worsening matching quality. Additionally, we control for time-varying housing supply constraints at the city level using the MSA-level Land Unavailability (LU) measure developed by Lutz and Sand (2023).²² The error term v_{it} therefore captures unobservable factors influencing house price growth persistence.

Eq. (4) is estimated using two panel data approaches: TWFE and CCE estimators. In panel data analysis, cross-sectional dependence must be addressed, as highlighted by Oikarinen et al. (2018). This dependence arises from factors such as mobility, trade (e.g., Choi et al., 2020), and multi-market bank networks (e.g., Choi and Hansz, 2021), which can create strong correlations in urban house price growth

¹⁹ Although loans are more directly tied to credit supply, we use deposits as a proxy for two reasons. First, banks often originate loans using deposit funding, suggesting that the availability of local deposits is closely linked to loan supply. This link is particularly strong for smaller banks with limited branch networks, which typically issue loans in the same regions where they collect deposits (e.g., Aguirregabiria et al., 2020). Becker (2007) also demonstrates that local deposit supply has a significant impact on the local banking sector and economy. Second, while HMDA loan data are well-suited for our analysis, they are only available from 2004 onward at the current MSA level following major revisions to MSA boundaries and updates to Regulation C by the Federal Reserve Board. Although not reported here for brevity, we find a positive relationship between per capita deposits and HMDA loan amounts across MSAs.

²⁰ Examining the effects of the Interstate Banking and Branching Efficiency Act (IBBEA) of 1994 on households' homeownership and home equity outcomes, Taşkın and Yaman (2023) show that the expansion of bank branching significantly increased mortgage supply, thereby stimulating housing demand and raising home prices. This effect was particularly strong in urban areas (Favara and Imbs, 2015).

²¹ As discussed further in the Online Appendix, we find that the estimated effect of the extrapolative expectations (EXP) variable depends on the selected lookback horizon: it is significantly positive when measured using the 2-year house price growth gap, but significantly negative when using the 4-year and 8-year gaps. This pattern indicates that the role of EXP measure is sensitive to the choice of time window.

²² Lutz and Sand (2023) extended the land availability measures developed by Saiz (2010) to finer geographic levels, creating the LU index, which measures the share of undevelopable land at the MSA level. Unlike many other housing supply constraint measures, the LU index varies both across MSAs and over time, covering 720 counties between 2002 and 2022. See Appendix C for further details on the LU index.

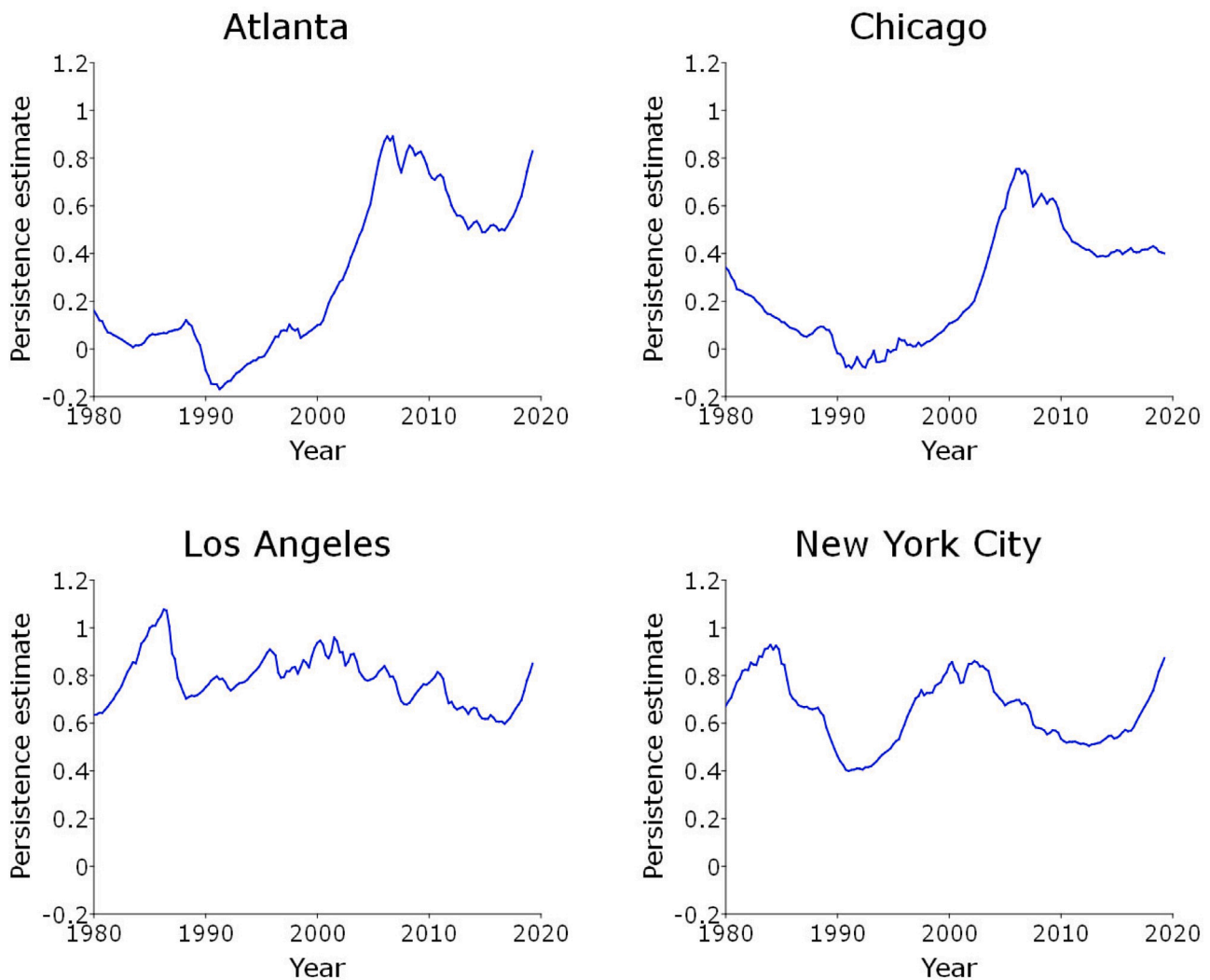


Fig. 7. Time-varying persistence estimated from TVP-SV-AR model for four selected MSAs.

Notes: This figure illustrates the persistence estimates ($\hat{\rho}_{it}$) from Fig. 6, focusing on four selected cities.

across cities. The two panel data techniques considered here differ in dealing with the cross-sectional dependence: unobserved common time-effects for the TWFE model (δ_i in Eq. (4)) and unobserved common factors with heterogeneous loadings (or interactive effects) in the CCE approach ($\gamma'_i f_t$ in Eq. (4)). The CCE method is generally more robust compared to TWFE. As a result, the CCE estimation method by Pesaran (2006) has become widely used, leading to an extensive literature that expands on the original modeling assumptions of Pesaran (2006) (see Westerlund and Kaddoura, 2022, and references cited therein for recent methodological advancements). We report both the CCE mean group (CCEMG) and CCE pooled (CCEP) estimators in our analysis.

Table 3 displays the results from our panel data regressions. Although the TWFE and CCE estimators yield slightly different magnitudes – likely reflecting their distinct treatments of cross-sectional dependence (Han et al., 2022) – the two key explanatory variables, CS and EXP , consistently exhibit the expected positive signs and remain statistically significant across panel estimation methods.²³ To

interpret, the regression results provide support for both the extrapolative expectations and credit supply channels.²⁴ For the control variables (W_{it}), we find mixed results. Per capita income growth consistently shows a positive and significant impact, while the coefficients for population density growth and the unemployment rate are statistically insignificant.

A similar pattern is observed in the lower panel of Table 3, where city-level Land Unavailability (LU) is included to account for time-varying housing supply constraints. Interestingly, the coefficient for housing supply constraints (LU) is consistently positive and statistically significant across panel methods employed. This result supports previous research indicating that house prices exhibit more persistent movements in cities with greater supply constraints (e.g., Capozza et al., 2004; Lai and Order, 2017; Oikarinen et al., 2018).

To sum, our panel data regression results indicate that both the credit supply channel and the extrapolative expectations channel help explain the observed persistence in urban house prices. In particular,

²³ More specifically, while the TWFE specification yields a positive but statistically insignificant coefficient on credit supply (CS), our inference is based on the CCE estimators. This choice is motivated by the CRT test of Castagnetti et al. (2015), reported at the bottom of Table 3, which strongly favors the CCE framework over the TWFE model for our data, as the test statistic (96.60) substantially exceeds the 5% critical value of 14.31.

²⁴ A caveat is warranted. As discussed in the Online Appendix, the estimated effect of EXP is sensitive to the choice of expectation horizon: while the coefficient exhibits the expected sign at shorter horizons, it turns negative, contrary to theoretical predictions, when the horizon exceeds two years. This represents an interesting avenue for future research but lies beyond the scope of the current paper.

Table 3

Results of panel data regression for two samples.

Explanatory variables	TWFE	CCEP	CCEMG
(A) Model 1: for 279 MSAs over 1994–2020			
CS (deposit growth)	0.015 [0.015]	0.032* [0.020]	0.298‡ [0.071]
EXP (2-yr Hamilton filtered)	0.148‡ [0.069]	0.516‡ [0.084]	0.298* [0.175]
Per capita income growth	0.192* [0.105]	0.399‡ [0.147]	0.932‡ [0.205]
Population density growth	0.041 [0.045]	0.178 [0.164]	4.706‡ [1.201]
Unemployment rate	0.011* [0.006]	0.015‡ [0.006]	0.012 [0.008]
Observations	7533	7533	7533
CRT test	Test-statistics: 96.60 Critical value: 14.31		
(B) Model 2: for 261 MSAs over 2002–2020			
CS (deposit growth)	0.013 [0.014]	0.043* [0.022]	0.307‡ [0.099]
EXP (2-yr Hamilton filtered)	−0.062 [0.075]	0.253‡ [0.113]	0.237 [0.270]
Per capita income growth	0.047 [0.106]	0.384‡ [0.141]	0.324 [0.259]
Population density growth	0.052 [0.043]	0.489* [0.275]	4.101‡ [1.341]
Unemployment rate	0.010* [0.006]	−0.002 [0.007]	−0.016 [0.015]
Land Unavailability (LU)	0.181‡ [0.049]	0.132‡ [0.077]	0.176* [0.104]
Observations	4959	4959	4959
CRT test	Test-statistics: 100.00 Critical value: 14.31		

Notes: The regression equation is Eq. (4), namely

$$\hat{\rho}_{it} = \mu_i + \beta_1 CS_{it} + \beta_2 EXP_{it} + W_{it}'\varphi + v_{it},$$

where the dependent variable is the time-varying persistence measure ($\hat{\rho}_{it}$) of MSA-level house price growth estimated in the TVP-SV-AR model. CS_{it} denotes the bank deposit growth as a proxy for credit supply measure, and EXP_{it} represents the previous two-year house price gap based on the Hamilton filter (2018). W_{it} is a vector of housing demand fundamentals, including per capita income, population density, and the unemployment rate, as well as the housing supply factor (LU) in Model 2. The total number of observations is 7533 for 279 MSAs during 1994–2020 for Model 1 and 4959 observations for 261 MSAs over 2002–2020. ‡, † and asterisk (*) respectively indicate the statistical significance at the 1%, 5%, and 10% significance levels with the corresponding clustered s.e. inside square brackets.

the findings suggest that backward-looking extrapolative expectations contribute to the persistence of house price growth, consistent with prior evidence (e.g., Glaeser and Nathanson, 2017; Kuchler et al., 2023). At the same time, the results align with a large literature identifying shifts in credit conditions as a key driver of housing market fluctuations (e.g., Favara and Imbs, 2015; Favilukis et al., 2017; Griffin et al., 2021; Mian and Sufi, 2022). Notably, the expansion of mortgage credit supply in the early 2000s is often attributed to the geographic expansion of bank branch networks following banking deregulation, which facilitated the diffusion of credit across state and regional boundaries. This expansion in mortgage lending capacity stimulated housing demand and accelerated price appreciation, particularly in historically credit-constrained regions such as the South and the Midwest.

5.2. Endogeneity concerns and an IV analysis

Our analyses in the preceding section suggest that both the credit supply channel and the extrapolative expectations channel can explain the house price persistence. Unlike the backward-looking extrapolative expectations channel, however, identifying the causal effects of credit supply expansion is challenging due to potential endogeneity concerns. One such concern is omitted variable bias, where unobserved factors may be correlated with both credit supply and house price growth persistence, resulting in $Cov(CS_{it}, \epsilon_{it}) \neq 0$ in Eq. (4).

While we mitigate this issue by incorporating additional time-varying control variables and allowing for unobserved common factors to correlate with regressors in the CCE approach, omitted variables could still simultaneously influence credit supply and house price persistence. Another endogeneity concern is reverse causality: persistent increases in house prices (and consequently in collateral values) might lead to greater credit availability and provision (e.g., Gelain and Lansing, 2014).

Table 4

IV regression results using city-level Bartik instrument.

Explanatory variables	TWFE	CCEP	CCEMG
CS (deposit growth)	0.299‡ [0.026]	0.118‡ [0.010]	0.100‡ [0.007]
EXP (2-yr Hamilton filtered)	7.931‡ [1.069]	20.613‡ [2.009]	17.794‡ [3.109]
Per capita income growth	0.371‡ [0.084]	0.340‡ [0.128]	0.774‡ [0.150]
Population density growth	0.051* [0.027]	−0.148 [0.116]	1.399 [0.910]
Unemployment rate	0.021‡ [0.004]	0.019‡ [0.006]	0.017‡ [0.007]

Notes: The regression equation is

$$\hat{\rho}_{it} = \mu_i + \beta_1 \widehat{CS}_{it} + \beta_2 EXP_{it} + W_{it}'\varphi + v_{it},$$

where $\widehat{CS}_{it}$ is the predicted values of CS_{it} estimated from $CS_{it} = \mu_i + \eta_1 D_{it} + \eta_2 EXP_{it} + W_{it}'\varphi + v_{it}$ where D_{it} is the city-level Bartik instruments constructed by $D_{it} = \sum \delta_{it} \cdot \Delta Dep_{-i,t}^{US}$. All the other variables are the same as in the footnote of Table 3.

To address these endogeneity concerns, we adopt an instrumental variable (IV) approach for credit supply, following established literature. Identifying valid and strong instruments is by no means straightforward, but we construct a Bartik (1991) type shift-share instrument for local credit supply (e.g., Borusyak et al., 2022). The Bartik (1991) type instrument projects national data onto local regions to ensure instrument exogeneity. Shift-share measures have been widely used as instruments for local house prices based on local exposure to exogenous national productivity trends (e.g., Mian et al., 2013; Saiz, 2010).

Our instrument is defined as:

$$D_{it} = \delta_{it} \cdot \Delta Dep_{-i,t}^{US}, \quad (5)$$

where δ_{it} represents the local deposit share in city i at time t , and $\Delta Dep_{-i,t}^{US}$ denotes the national deposit growth at time t excluding the city i . National deposit growth is computed as a leave-one-out measure to avoid mechanical correlation between the national trend and city i 's credit supply (Borusyak et al., 2022). Thus, D_{it} serves as a city-level Bartik-type “shift-share” instrument for local credit supply, capturing exogenous local credit supply changes through the interaction of local deposit share (δ_{it}) and national deposit growth ($\Delta Dep_{-i,t}^{US}$).

Table 4 presents the IV regression results using the city-level Bartik instruments (D_{it}) for credit supply in panel data regressions of Eq. (4). Despite some variations in specific cases, the results are largely consistent with those in Table 3. They demonstrate that the impacts of both credit supply and extrapolative expectations remain consistently positive and significant across all models. This confirms the robustness of our findings, even when addressing endogeneity with instrumental variables.

5.3. The role of housing supply constraints

So far, our focus has been on how housing demand factors contribute to variations in persistence over time. However, some studies emphasize the role of housing supply factors, such as regulatory constraints and associated costs, in driving fluctuations in house prices (e.g., Aastveit et al., 2023a; Capozza et al., 2004; Glaeser et al., 2008; Huang and Tang, 2012; Saiz, 2010). Moreover, empirical evidence suggests that the impact of housing demand factors on persistence is influenced by the availability of housing supply in local markets (e.g., Aastveit et al., 2023a; Capozza et al., 2004; Duca et al., 2021; Lai and Order, 2017; Oikarinen et al., 2018). Regulatory constraints and geographical limitations can restrict the housing supply's responsiveness to demand changes, thereby increasing house price persistence. Huang and Tang (2012) show that more restrictive housing supply constraints lead to larger responses of housing prices to economic shocks, resulting in greater persistence. Similarly, Aastveit et al. (2023a) note that varying housing supply elasticities contribute significantly to cross-city differences in house price movements. Given this context, it is worth exploring whether the effects of credit supply and extrapolative expectations on house price growth persistence vary depending on

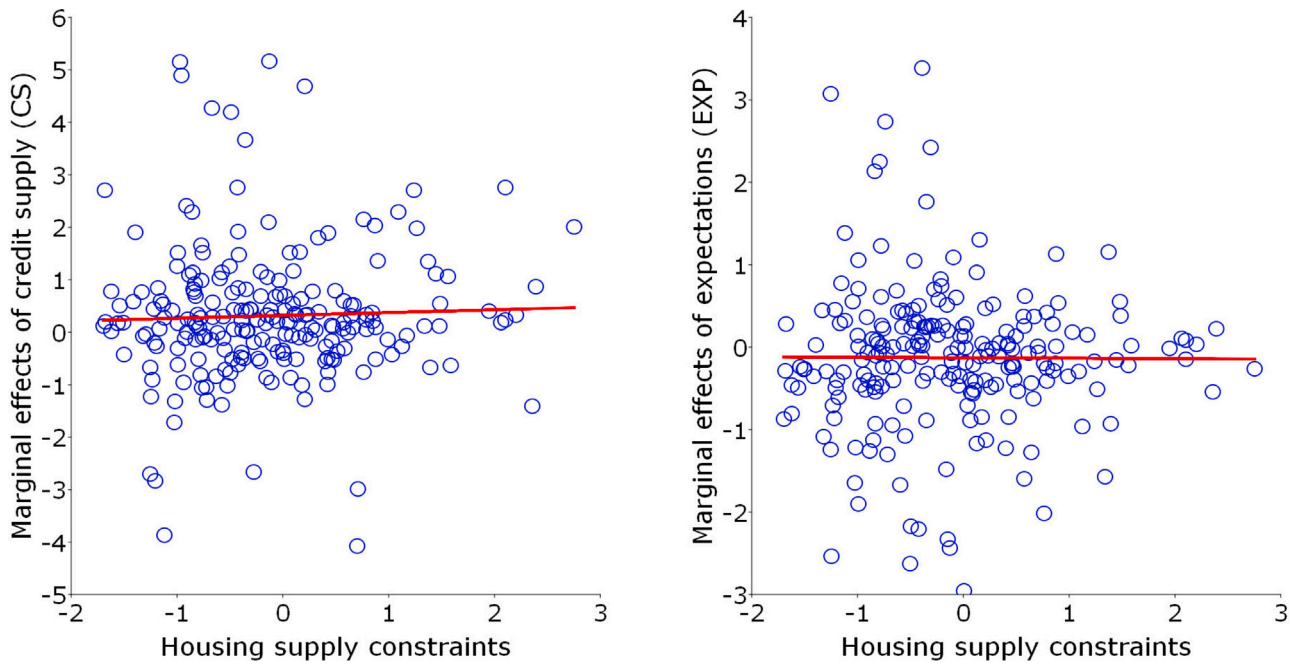


Fig. 8. Relationship between housing supply constraint measures (on the horizontal axis) and marginal effects of housing demand factors on persistence (on the vertical axis).

Notes: This figure illustrates the relationship between the WRLURI (a measure of housing supply constraints) and the marginal effects of credit supply (left panel) and expectations (right panel) on time-varying persistence.

housing supply constraints. Intuitively, a stronger effect of housing demand factors is anticipated in cities with stricter housing supply constraints.

To examine this, we first use the Wharton Residential Land Use Regulation Index (WRLURI) as a measure of housing supply constraints at the city level. This index, developed by Gyourko et al. (2008) and updated by Gyourko et al. (2021), provides a summary measure of the stringency of local regulations concerning permitting, zoning, and entitlement processes. A higher value of WRLURI indicates a more restrictive regulatory environment. Previous research has shown that cities with stricter housing regulations tend to experience more rapid and volatile house price movements (e.g., Choi and Hansz, 2021; Gyourko et al., 2008).²⁵

We start by analyzing the CCEMG estimation results from the previous section to investigate the relationship between the MSA-specific marginal effects of credit supply ($\hat{\beta}_{1,i}$) and expectations ($\hat{\beta}_{2,i}$) and the degree of housing supply constraints across 279 MSAs. If the impact of credit supply or future house price expectations is stronger in cities with more stringent housing supply constraints, we would expect to see a positive relationship. As shown in Fig. 8, however, there appears to be no clear correlation between the WRLURI and the marginal impacts of credit supply (left panel) and expectations (right panel), as indicated by the nearly horizontal fitted lines. This suggests that the influence of credit supply or expectations on persistence may not necessarily be stronger in cities with stricter housing supply constraints. This finding aligns with recent research by Howard and Liebersohn (2023), which indicates that housing supply elasticity alone does not account for much of the variation in house price appreciation, especially in less regulated regions. It also corresponds with our earlier observation that regions with relatively lax housing supply constraints, such as the Midwest and South, experienced a more notable increase in persistence.

²⁵ Although not reported here to conserve space, our results are largely similar using alternative popular measures of housing supply constraints, such as the land unavailability measure of Saiz (2010) and the inverse of the housing supply elasticity by Guren et al. (2021).

Table 5

The role of housing supply constraints.

Explanatory variables	TWFE	CCEP	CCEMG
D_{it}	0.010 [0.023]	0.443‡ [0.189]	6.011* [3.539]
$D_{it} \times LU_{it}$	−0.036 [0.064]	−0.293 [0.417]	−8.527 [9.505]
Per capita income growth	0.067 [0.106]	0.089 [0.122]	0.019 [0.274]
Population density growth	0.037 [0.043]	0.038 [0.156]	1.489 [1.252]
Unemployment rate	0.009 [0.005]	0.000 [0.006]	0.009 [0.012]

Notes: The regression equation is

$$\hat{\rho}_{it} = \alpha D_{it} + \beta(D_{it} \times LU_{it}) + W'_{it}\Phi + \mu_i + z_i + \varepsilon_{it},$$

where D_{it} is the city-level Bartik shift-share instrument for the exogenous credit supply shock, constructed by $D_{it} = \sum \delta_{i,t_0} \cdot \Delta Dep_{-i,t}^{US}$. LU_{it} is the measure of Land Unavailability in city i at time t . All the other variables are the same as in the footnote of Table 3.

Additionally, we run the following panel data regression,

$$\hat{\rho}_{it} = \alpha D_{it} + \beta(D_{it} \times LU_{it}) + W'_{it}\Phi + \mu_i + z_i + \varepsilon_{it}, \quad (6)$$

where D_{it} denotes the city-level exogenous credit supply shock defined in Eq. (5), and LU_{it} represents the measure of Land Unavailability in city i at time t . The other variables are the same as those in Eq. (4). The interactive term ($D_{it} \times LU_{it}$) captures whether the response of persistence to local credit supply shock varies with the degree of local housing supply constraints. Given the well-established positive effect of housing supply constraints on house price growth persistence (e.g., Aastveit et al., 2023a), a positive sign is anticipated for the coefficient β . The results, presented in Table 5, show no significant effect of the interaction term ($D_{it} \times LU_{it}$). This suggests that the influence of credit supply on house price persistence does not depend on the local housing supply constraints.

Taken together, we find no strong evidence that the impact of credit supply or expectations on house price persistence is amplified in cities with stricter housing supply constraints. This finding is in line with the suggestive evidence in Louie et al. (2025), who indicate that supply constraints may have only a limited role in explaining cross-city variation in rising house prices. However, this does not necessarily

refute the positive effect of housing supply constraints *per se* on house price persistence, as shown in Table 3(B) as well as in prior studies (e.g., Lai and Order, 2017; Oikarinen et al., 2018). Instead, our results imply that the effect of credit supply or extrapolative expectations on house price persistence may operate through mechanisms other than housing supply constraints.

6. Concluding remarks

The persistence of house price growth has been a salient feature of the U.S. housing markets. Despite its significant implications on key economic areas such as household consumption and the labor market, relatively little is known about the stability of persistence over time, let alone the factors driving it. Understanding the time-varying feature of house price persistence is critical, given the prominent role of housing in the broader economy. Policymakers, in particular, could benefit from insights into the factors that influence temporal shifts in persistence. The substantial geographic disparities in house price persistence across U.S. regions suggest that U.S. monetary policy may have uneven regional effects.

We take a novel approach to estimating house price persistence. By employing a Time-Varying Parameter Stochastic Volatility Autoregressive (TVP-SV-AR) model, we provide compelling evidence of time variation in persistence of national and urban house price growth, especially pronounced after the mid-1990s. The variation in persistence differs geographically, with regions previously exhibiting lower persistence experiencing greater fluctuations over time. This pattern cannot be fully explained by traditional housing demand or supply factors, suggesting the presence of additional factors influencing the patterns of house price persistence.

To better understand these drivers, we focused on two key housing demand factors frequently highlighted in recent research: credit supply and extrapolative expectations. By linking geographic differences in persistence to city-level characteristics through panel data regressions, we find that both credit supply and expectations can account for the time-varying nature of persistence, although the expectations channel is relevant only over short expectation horizons, typically within two years. In particular, the increased persistence of house price growth in areas with limited capital, such as the Midwest and South, appears to be driven by the expansion of credit facilitated by interstate banking integration and securitization. Although house price persistence is also positively correlated with stricter housing supply constraints, as often documented in the literature, the effect of credit supply on persistence does not seem to depend on the severity of these constraints. Our findings remain robust after addressing potential endogeneity using a Bartik type shift-share instrumental variable.

While this study offers new insights into U.S. house price movements, it also underscores the need for further research. Future studies should explore the time-varying behavior of house price persistence across future housing cycles to gain a more comprehensive understanding of its dynamics. Moreover, since our analysis relies on aggregate city-level data, more granular investigations are necessary to mitigate potential aggregation bias. Exploring other sources of geographic variation in persistence beyond the two housing demand factors discussed here could also provide valuable insights. Nevertheless, our study makes distinctive contributions by highlighting that house price growth persistence is far from stable, and further exploration is needed to gain a better understanding of its evolving nature.

CRedit authorship contribution statement

Chi-Young Choi: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alexander Chudik:** Methodology, Formal analysis, Conceptualization. **Aaron Smallwood:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation.

Appendix A. Sum of autoregressive coefficients (SARC) in AR model

A number of approaches have been employed to assess the extent of the persistence in house price growth. One such approach is to use a univariate autoregressive (AR) time series model. We measure the persistence of each housing price growth rate using the sum of autoregressive coefficients (SARC) in the AR(p) representation given by

$$y_t = \alpha + \sum_{j=1}^p \beta_j y_{t-j} + \varepsilon_t$$

$$= \alpha + \rho y_{t-1} + \sum_{k=1}^{p-1} \zeta_k \Delta y_{t-k} + \varepsilon_t,$$

where y_t represents the growth rate of house price at time t and $\zeta_k = -\sum_{j=k+1}^p \beta_j$. As such, $\rho = \sum_{j=1}^p \beta_j$ denotes the SARC. The lag length (p) is selected using the BIC with a maximum lag length of 8. To deal with the well-known downward small sample bias embedded in the OLS estimation of ρ (Choi et al., 2006), we follow the common practice in previous studies (e.g. Choi and O'Sullivan, 2013) and employ the Hansen's (1999) 'grid bootstrap' based median-unbiased (MUB) estimator. Persistence is higher for a larger value of the SARC (ρ).

Appendix B. Measures of extrapolative expectations

As a proxy for extrapolative expectations, we use the house price gap, defined as the deviation of observed house price growth from its long-run trend. The trend component is estimated using the regression-based filter proposed by Hamilton (2018). Specifically, the Hamilton filter decomposes house price growth (y_{t+h}) into a predictable component, $E[y_{t+h}|y_t, y_{t-1}, y_{t-2}, y_{t-3}]$ (interpreted as the trend), and a forecast error v_{t+h} , which represents the cyclical component:

$$y_{t+h} = \alpha + \beta_0 y_t + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \beta_3 y_{t-3} + v_{t+h},$$

where h denotes the forecast horizon, i.e., the number of periods ahead for which house price growth is predicted using lagged information. The estimated residual $\hat{v}_{t+h}$ represents the house price growth gap, capturing cyclical fluctuations in house price growth relative to its longer-run trend.

The basic idea is that expectations about future house price growth are shaped by departures from long-run trends rather than by short-run fluctuations alone. Large deviations from trend characterize environments in which recent price growth is unusually strong and salient, increasing the likelihood that past appreciation is extrapolated into the future. For this reason, the house price gap provides a more informative proxy for extrapolative expectations than simple measures of lagged house price growth.

In the existing literature, the house price gap is often constructed as the deviation of log house prices from their Hodrick–Prescott (HP) trend. However, Hamilton (2018) raises several concerns regarding the reliability of HP-filter-based estimates, including spurious dynamic behavior, sensitivity to arbitrary smoothing parameters, and end-of-sample bias. When present, these issues may distort the true cyclical position of house prices and thereby compromise the accuracy of the resulting gap measures. As an alternative, Hamilton (2018) proposes a regression-based filter that addresses these shortcomings while preserving the objectives motivating the use of the HP filter. Unlike band-pass filters such as the HP filter, the Hamilton filter relies on a concept of cycles that is closely related to the Beveridge–Nelson (1981) decomposition.

As discussed in the Online Appendix, we find that the regression results are sensitive to the choice of forecast horizon (h).

Appendix C. Measures of housing supply constraints

The Wharton Residential Land Use Regulation Index (WRLURI), developed by Gyourko et al. (2008) and subsequently updated by Gyourko et al. (2021), is a popularly used measure of the stringency of local housing regulations. It comprises eleven sub-indexes that provide a summary of various aspects of the regulatory environment. A low value of the WRLURI index indicates a less restrictive or more permissive approach to regulating the local housing market. A large body of research (e.g., Choi and Hansz, 2021; Gyourko et al., 2008) has shown that cities with more stringent housing regulations, and hence lower elasticity of housing supply, experience faster and more volatile growth in house price. As such, the WRLURI index provides a valuable tool for measuring the stringency of local housing regulations and their impact on the dynamics of house prices in different cities. Our results are largely similar using alternative popular measures of housing supply constraints, such as the land unavailability measure of Saiz (2010) and the inverse of the housing supply elasticity by Guren et al. (2021).

Another measure of housing supply constraints considered in the current study is the Land Unavailability (LU) measure developed by Lutz and Sand (2023). The basic idea of the LU measure is that the amount of land available for development plays a crucial role in local housing supply elasticity. While other measures of housing supply constraints, such as the WRLURI, solely capture cross-sectional variation, the LU measure varies not only across geographic units but also over time. Specifically, it spans 720 counties from 2002 to 2022 and 1828 counties from 2011 to 2022, making it suitable for our panel data regression analysis as presented in Table 3. As noted by Conklin et al. (2022), the LU index is highly correlated with the Saiz's (2010) measure.

Appendix D. Comparison between the TWFE model and the CCE estimation

The basic idea of the TWFE model is to estimate the marginal effect of the explanatory variables of interest, while taking into account city-level heterogeneity (μ_i) and common macroeconomic shocks (δ_t). The MSA-fixed effects (μ_i) control for potentially confounding omitted variables, including locational differences in economic growth, housing supply elasticity, and demographic trends. The year-fixed effects (δ_t) control for year-specific unobserved confounders that are common to all cities, such as macroeconomic shocks. The inclusion of year-fixed effects is particularly important because house prices may move in different cities due to national house price trends, such as changes in mortgage rates or credit supply changes. By controlling for these national trends, the TWFE model permits us to isolate the effects of the variables of interest on persistence.

The TWFE model, however, assumes that the error terms (ϵ_{it}) in Eq. (4) are uncorrelated with the explanatory variables, which may not hold if unobserved common factors exist that cause non-zero correlations between explanatory variables and error terms. Moreover, if common nationwide shocks, such as monetary policy shocks, affect all MSAs to varying degrees, the year-fixed effects (δ_t) in the TWFE model may not be able to capture the heterogeneous effects of common shocks (Han et al., 2022). To address this issue, Pesaran (2006) proposes the use of Common Correlated Effects (CCE) estimators. The basic idea behind CCE estimators is to control for unobserved common factors by augmenting the model with cross-sectional averages of the dependent variable as well as explanatory variables. CCE estimation is known to be robust to different types of cross-sectional dependence in the error term, possible unit roots in factors, and slope heterogeneity.

Appendix E. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbankfin.2026.107749>.

Data availability

Data will be made available on request.

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